Web of Science Practice

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Question 1. *Provide the citation for a review article that cites Serlin, Tschirhart, Polshyn, Zhang, Zhu, Watanabe, Taniguchi, Balents and Young, "Intrinsic quantized anomalous Hall effect in a moiré heterostructure" Science 367:900-903 (2020). https://doi.org/10.1126/science.aay5533*

Proof. Here is one:

Li, Redekop, Beach, Zhang, Zhang, Liu, Holtzmann, Hu, Anderson, Park, Taniguchi, Watanabe, Chu, Fu, Cao, Xiao, Young and Xu, "Universal Magnetic Phases in Twisted Bilayer $MoTe_2$" Nano Letters 25:18044-18050 (2025). https://pubs.acs.org/doi/10.1021/acs.nanolett.5c04751 □

Question 2. *How many publications are referenced in the most cited paper co-authored by Vera Kistiakowsky?*

Proof. There are 23 publications referenced in the most cited paper co-authored by Vera Kistiakowsky. □

Question 3. *How many peer-reviewed articles cite Cornell & Wieman's first experimental report of trapping a Bose-Einstein condensate.*
Check that:

- *your answer counts only peer-reviewed articles.*
- *the report describes an experiment, not just its results (read the abstract, or the entire content).*
- *The report does not reference any earlier work on the same topic by the same authors.*

Proof. There are total of 7 peer-reviewed articles that cite Cornell & Wieman's first experimental report of trapping a Bose-Einstein condensate (while also providing description of some experimentm and not referencing other early work by Cornell & Wieman). □

Question 4. *Find the oldest article in the American Journal of Physics measuring the speed of light.*

a. Proide its citation.

b. Describe its measurement approach.

Proof. The oldest article in AJP I found that's about measuring the speed of light, is "A laboratory experiment on the velocity of light" in 1954.

a. Citation:

Palmer, Spratt, "A laboratory experiment on the velocity of light." Am. J. Phys. 22:481-485 (1954). https://doi.org/10.1119/1.1933791

b. The measurement of speed of light here requires a light source, and a Kerr cell shutter to produce light pulses. Then, they used a half-silvered mirror to divide the light pulses into two beams (in orthogonal directions). With the two beams traveling different distances, let them get reflected back, and combined to be measured by the deterctor. Then, by adjusting the distance each beam travels, knowing the frequency of the light pulses, and teh light's behavior detected (as sinusoidal waves), these data can be used to directly calculate the speed of light.

□

Question 5. *Find Bednorz & Muller's first experimental report of high-Tc superconductivity.*

a. Provide its citation.

b. Which UCSB Physics professor most recently authored an article that cites it?

c. What year was the oldest article it cites published?

Proof. a. The most cited article regarding the first experimental report of high-Tc superconductivity by Bednorz & Muller is: "Possible high-Tc superconductivity in the Ba-La-Cu-O system" in 1986.

Citation:

Bednorz, Muller, "Possible high-Tcj superconductivity in the Ba-La-Cu-O system." Zetschrift fur Physic B-Condensed Mattter 64:189-193 (1986). https://doi.org/10.1007/BF01303701

b. Leon Balents is the UCSB Physics professors most recently authored an article that cites the mentioned paper. The article is: "Superconductivity and strong correlations in moire flat bands" in 2020.

c. The oldest (avaiabl on Web of Science) article it cites is from 1973, with title "High-temperature superconductivity in Li-Tl-O Ternary-system".

□

Question 6. *Find the most recent article co-authored by UCSB's Chancellor Henry T Yang.*

a. Provide its citation.

b. How many pages long is the article?

Proof. a. The most recent article co-authored by Chancellor Yang is: "A generalized load-penetration relation for sharp indenters and the indentation size effect" in 2002.

Citation

Li, Chandrasekar, Yang, "A generalized load-penetration relation for sharp indenters and the indentation size effect." Journals of applied mechanics-transactions of the ASME, 69:394-396 (2002). https://doi.org/10.1115/1.1458557

b. The paper is 3 pages long.

□

Question 7. *Find the most cited article co-authored by a member of the UCSB Physics Department.*

a. Provide its citation.

b. How many papers cite it?

c. How many papers does it cite?

d. What is the e-mail address of the corresponding author of the most recent article that cites it?

Proof. The member I chose is Professor Andrea Young.

a. The most cited article that he has co-aurthored is: "Tuning superconductivity in twisted bilayer graphene" in 2019.

Citation:

Yankkowitz, Chen, Polshyn, Zhang, Watanabe, Taniguchi, Graf, Young, Dean, "Tuning superconductivity in twisted bilayer graphene." Science 363:1059-1064. https://www.science.org/doi/10.1126/science.aav1910

b. There are 2067 papers that cite this one.

c. This paper cites 62 papers.

d. The mos recent article that cites it is: "Magic angle effect driven by deep learning: Giant electromechanica coupling in four-layer twisted lithium niobite" indexed in December 2025, scheduled to publish in July 2026.

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